Q-Engine 2026: Superconducting Quantum Processor Architecture

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1. System Overview and Cryogenic Parameters

The Q-Engine 2026 is a 256-qubit superconducting quantum processor utilizing planar transmon architectures. The dilution refrigerator maintains a base plate operating temperature of **14.5 milliKelvin (mK)** with a cooling power of 18 uW at base. Qubit coherence metrics demonstrate a longitudinal relaxation time (T1) of **185 microseconds** and a dephasing time (T2) of **242 microseconds**.

2. Gate Fidelity Benchmark Matrix

Processor Model	Qubit Count	Single-Qubit Fidelity	Two-Qubit (CZ) Fidelity	Readout Error
Q-Engine 128 (Gen 1)	128 Transmons	99.92%	99.45%	0.85%
Q-Engine 256 (Gen 2)	256 Transmons	99.98%	99.86%	0.32%
Q-Engine 512 (Preview)	512 Transmons	99.95%	99.78%	0.48%

3. Quantum Error Mitigation (QEM)

Real-time zero-noise extrapolation (ZNE) combined with randomized compiling suppresses coherent control errors by **84.2%**. The active surface code cycle time is clocked at **200 nanoseconds**.